

1. Answer: A

$$\text{Given } T = 2\pi\sqrt{\frac{L}{g}}$$

$$g = \frac{4\pi^2 L}{T^2}$$

$$\left(\frac{\Delta g}{g}\right) \times 100\% = \left(\frac{\Delta L}{L}\right) \times 100\% + 2\left(\frac{\Delta T}{T}\right) \times 100\%$$

$$2\% = \left(\frac{0.05}{6.25}\right) \times 100\% + 2\left(\frac{\Delta T}{T}\right) \times 100\%$$

$$\left(\frac{\Delta T}{T}\right) \times 100\% = 0.6\%$$

2. Answer: A